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Experiment 7: Schematic design of a Two-stage differential amplifier

Abstract

This experiment presents the schematic design and Cadence Virtuoso simulation of a two-stage CMOS differential amplifier. The amplifier was designed for a 1 pF load capacitor, a target phase margin near 60° in buffer configuration, and specified quiescent currents of 200 μA for the first stage and 800 μA for the second stage. The design was implemented with all transistor lengths fixed at 0.36 μm . Simulation results show a low-frequency open-loop gain of 56 dB, unity-gain frequency of 117.824 MHz, buffer phase margin of 59.172° , slew rate of 187.213 V/ μs , settling time of 1.011 44 μs , input common-mode range of 0.955 V to 1.505 V, output swing of 0.298 V to 1.501 V, and power dissipation of 2.078 mW at 1.8 V supply and 1.2 V input common-mode DC bias.

1 Introduction

A two-stage differential amplifier is widely used in analog integrated circuits because it combines high gain with a large output swing. The first stage provides differential-to-single-ended conversion and most of the open-loop gain, while the second stage provides additional gain and output drive capability. In this experiment, the amplifier was designed and simulated in Cadence to satisfy the given specifications for gain, stability, input common-mode range, output swing, and power dissipation.

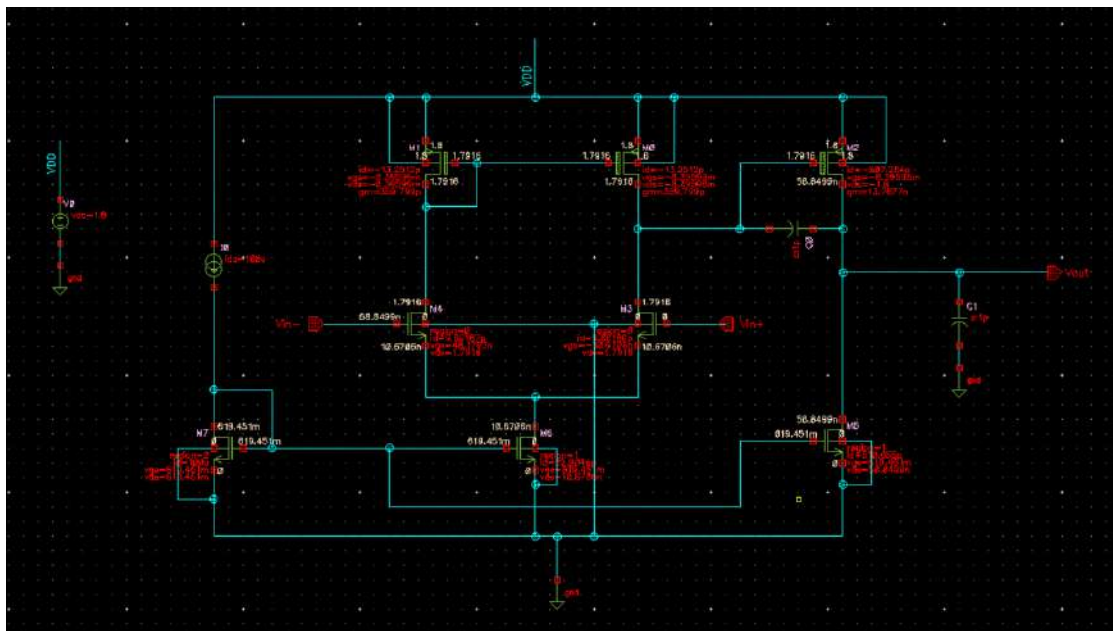
2 Design Specifications

The following design targets were used:

- Load capacitance: 1 pF
- Phase margin in buffer configuration: about 60°

- ### 3 Circuit Topology

3.1 Op-Amp Schematic



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3.2 Buffer Configuration Schematic

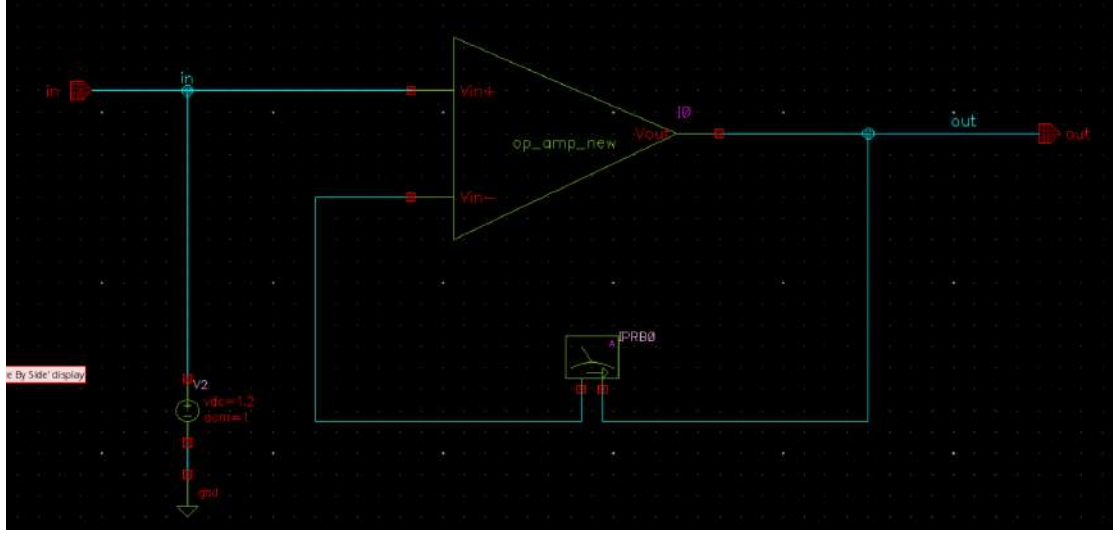


Figure 2: Schematic of the op-amp in buffer configuration.

3.3 Circuit Operation

- The differential input pair converts the input voltage difference into a current difference.
- The current mirror load converts the differential current into a single-ended voltage gain.
- The second stage boosts the voltage gain and drives the load capacitor.
- Miller compensation improves stability by creating a dominant pole at the first-stage output.

4 Device Sizing

All transistors were designed with channel length fixed at $0.36\mu\text{m}$. The widths used in the final design are listed below.

Table 1: MOSFET Dimensions Used in the Final Design

Device	Length (μm)	Width (μm)
NMOS differential pair	0.36	6
PMOS active load pair	0.36	9
Tail NMOS current source	0.36	18
Second-stage PMOS transistor	0.36	81
Second-stage NMOS transistor	0.36	72
NMOS current mirror device	0.36	9

5 Design Steps

The amplifier was designed and verified using the following procedure:

1. The supply voltage was fixed at 1.8 V.
2. The first stage was sized to satisfy the input common-mode range and the target quiescent current of 200 μA .
3. The second stage was sized to provide enough output drive and the target quiescent current of 800 μA .
4. A Miller capacitor of 1 pF was added for frequency compensation.
5. Open-loop AC analysis was performed to extract gain, unity-gain frequency, and phase margin.
6. The amplifier was then connected in buffer configuration, and the closed-loop stability was verified.
7. DC sweep analyses were performed to determine the input common-mode range and output swing.
8. Transient analysis was used to measure slew rate and settling time.
9. Power dissipation was measured from the supply current.

6 Simulation Results

The main simulated performance parameters are summarized in Table 2.

Table 2: Simulated Performance of the Two-Stage Differential Amplifier

Parameter	Simulated Value
Low-frequency open-loop gain	56 dB
Unity-gain frequency	117.824 MHz
Phase margin in buffer configuration	59.172°
Slew rate	187.213 V/ μ s
Settling time for 10% accuracy	1.01144 μ s
Input common-mode range (experimental)	0.955 V to 1.505 V
Output swing (experimental)	0.298 V to 1.501 V
Power dissipation at 1.2 V common mode DC bias	2.078 mW

7 Waveforms and Plots

7.1 Input Common-Mode Range Measurement

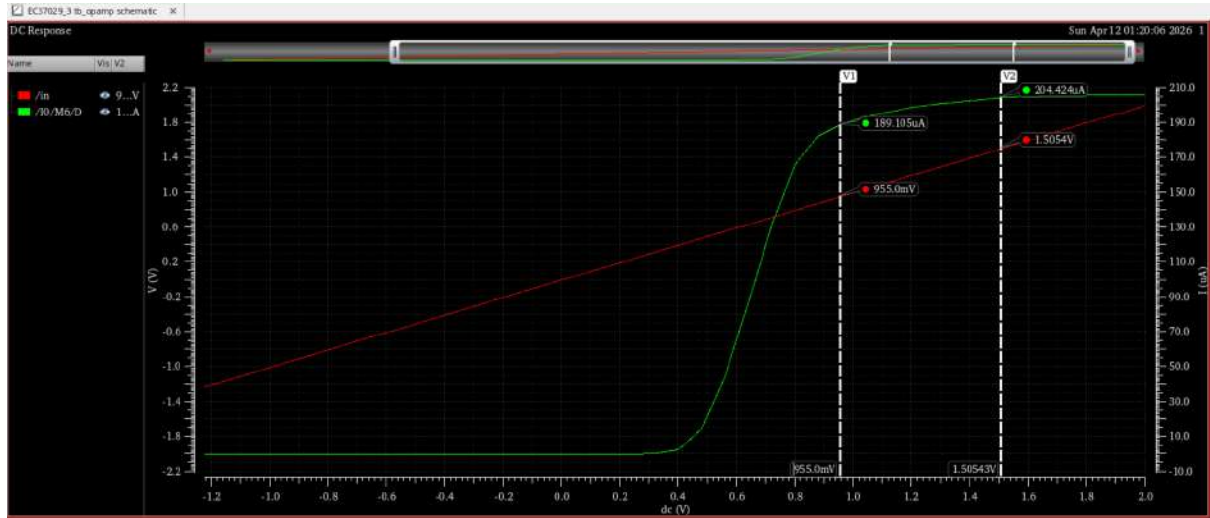


Figure 3: VDC sweep and corresponding tail current used to measure input common-mode range in buffer configuration.

7.2 Open-Loop Frequency Response



Figure 4: Open-loop frequency response showing gain and phase.

7.3 Closed-Loop Buffer Frequency Response

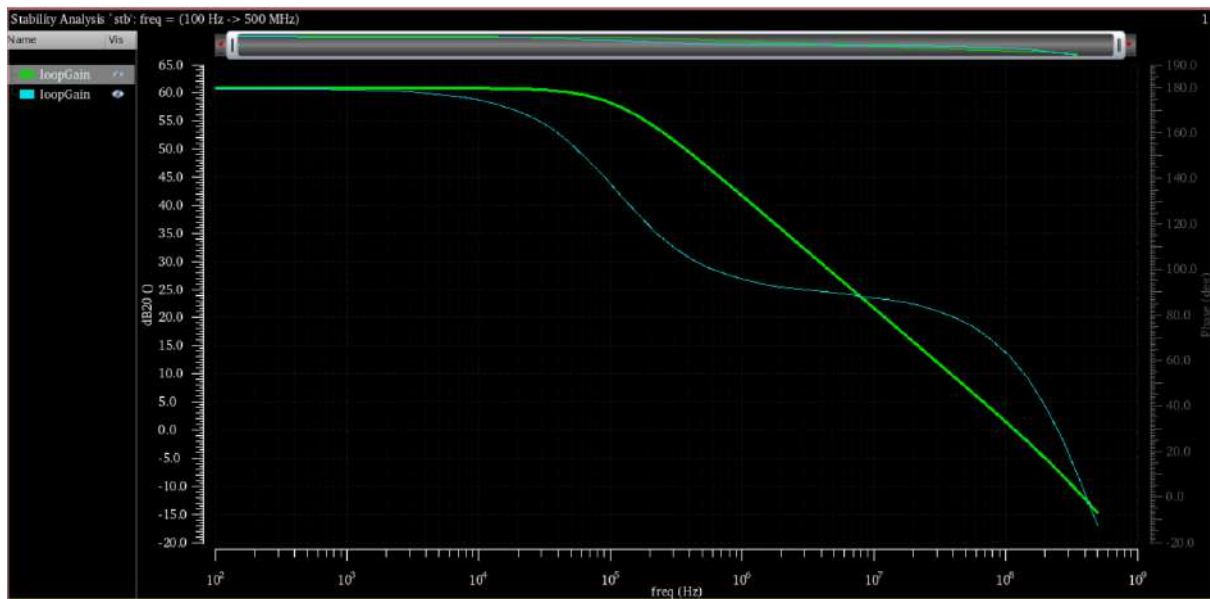


Figure 5: Closed-loop frequency response in buffer configuration showing gain and phase.

7.4 Output Swing Measurement

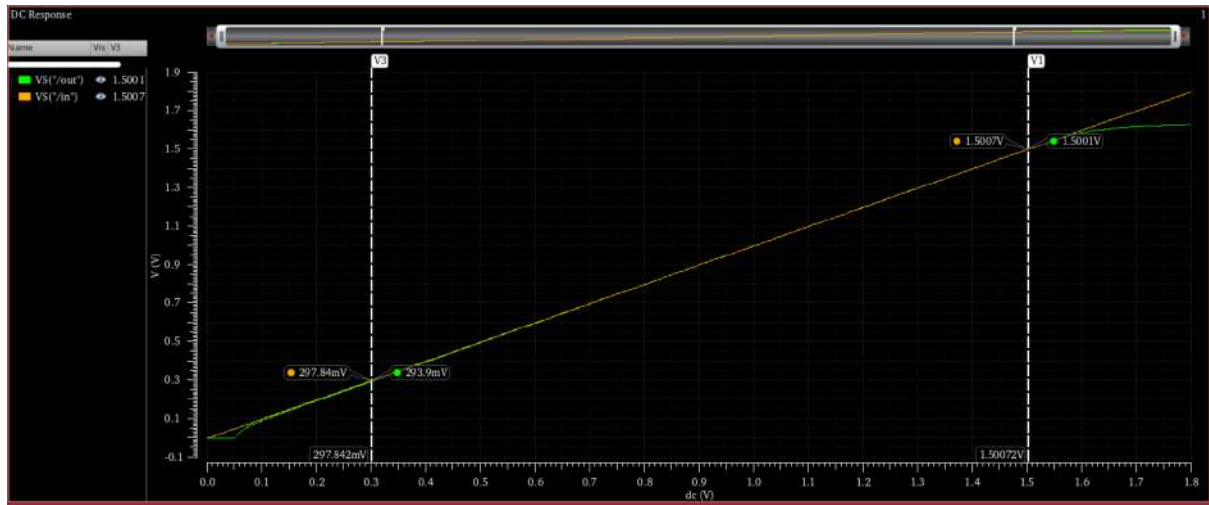


Figure 6: VDC sweep of input and corresponding output used to measure output swing in buffer configuration.

7.5 Stability Analysis

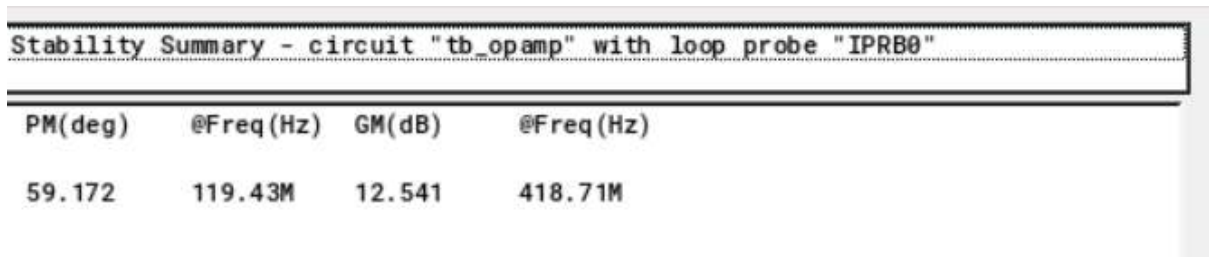


Figure 7: Stability analysis result of the buffer showing phase margin and gain margin.

8 Discussion

- The simulated low-frequency open-loop gain of 56 dB indicates satisfactory voltage amplification for a conventional two-stage CMOS operational amplifier.
- The unity gain bandwidth frequency 117.824 MHz shows that the amplifier exhibits high bandwidth as well as being stable at nearly the specified value.
- The phase margin found in the buffer mode is equal to 59.172° . Thus, it comes close to the ideal value 60° , proving that frequency compensation is correct and is done using the Miller capacitor with a capacitance of 1 pF.

- The gained input common mode range from 0.955 V to 1.505 V proves that there are no problems with the range of operating voltages for the input stage of the amplifier.
- The voltage output swing from 0.298 V to 1.501 V means that the amplifier can work close to the power supplies, which makes the amplifier appropriate for use in low-voltage analog circuits.
- The slew rate value of 187.213 V/ μ s proves that the amplification has a large signal fast-responding ability and depends on the bias current of the second stage and a compensation capacitor.
- The settling time of about 1.011 44 μ s 10% accuracy shows stable closed-loop performance with satisfactory transient response.
- The use of Miller compensation successfully ensured stability while introducing the expected trade-off between bandwidth and phase margin, which is typical in two-stage op-amp design.
- Overall, the designed amplifier satisfies the given specifications for gain, stability, bandwidth, input common-mode range, output swing, slew rate, and power consumption.

9 Conclusion

A two-stage CMOS differential amplifier was successfully designed and simulated in Cadence Virtuoso. The final circuit met the major design requirements for gain, stability, input common-mode range, output swing, and power dissipation. The amplifier achieved 56 dB open-loop gain, 117.824 MHz unity-gain frequency, and 59.172° phase margin in buffer configuration. The work demonstrates the standard trade-offs in analog IC design between gain, speed, stability, and power.